

Research on the impact of boundary-spanning search on the sustainable development ability of service-oriented manufacturing enterprises: evidence from Chinese listed companies

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Abstract

Purpose – This paper aims to analyze the mechanism underlying the impact of boundary-spanning search (BS) on the sustainable development ability (SDA) of service-oriented manufacturing enterprises and to emphasize the intermediary role of knowledge integration (KI). The moderating role of knowledge inertia on the link between BS and KI is also investigated.

Design/methodology/approach – This study constructs direct, mediating and moderating effects, selects 110 service-oriented manufacturing enterprises as research samples and obtains empirical data from questionnaires and annual reports. Among them, triangulation is skilfully used to obtain questionnaire data, and the regression method is used to test model relationships.

Findings – The results show that BS not only directly enhances SDA but also indirectly affects it through KI, which plays a mediating role in the impact of BS on SDA. In addition, knowledge inertia negatively moderates the relationship between BS and KI.

Originality/value – This paper makes three contributions. First, it enriches the research on the antecedent variables related to the SDA of service-oriented manufacturing enterprises. Second, by examining the mediating role of KI and the moderating role of knowledge inertia, the relationship between BS and SDA is revealed. Third, the research on knowledge management related to the SDA of service-oriented manufacturing enterprises is expanded.

Keywords Service-oriented manufacturing, Sustainable development ability, Boundary-spanning search, Knowledge integration, Knowledge inertia

Paper type Research paper

1. Introduction

The manufacturing industry is a pillar of China's national economy. China has a complete industrial chain; however, its manufacturing sector still faces problems, such as the low value-added of products and the homogenization of products. Therefore, China has established a number of policies, among which the *Made in China 2025* and the *Special Action Guide for the Development of Service-oriented Manufacturing* take the service manufacturing industry as an important strategic direction (Zhu *et al.*, 2021). Service-oriented manufacturing is an advanced manufacturing mode that integrates traditional manufacturing with servitization (Zhen, 2012). As a new sustainable production mode for manufacturing enterprises, it helps improve economic and environmental performance for promoting the circular economy (Seclen-Luna *et al.*, 2021), so that enterprises can maintain sustainable



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profitability and innovation ability in the development process, gain competitive advantage and achieve sustainable growth, which is SDA.

Many enterprises have improved the quality of products and services through service-oriented manufacturing. Taking the Yangtze River Delta region as an example of a developed area of the manufacturing industry and scientific and technological service industry, it gains benefits to promote regional service. It is a highly concentrated manufacturing area of China's service-oriented strategy (Lafuente *et al.*, 2017). However, some studies show that the overall level of China's service-oriented manufacturing industry is still low, and its service level is still at the low level of providing basic products and services and lacks high value-added services (Lerch and Gotsch, 2015; Wang *et al.*, 2017), which makes it difficult to improve SDA. Improving the SDA of service-oriented manufacturing enterprises is the greatest challenge for enterprises to implement service-orientation. In academia, scholars mainly examine the impacts on the SDA of enterprises from the perspectives of external economic stability (Pieloch-Babiarz *et al.*, 2021), stakeholder pressure (Seroka-Stolka and Fijorek, 2020) and so on, but these perspectives do not involve resource factors. The resource-based view points out that rich resources are conducive to the formation of imitable capabilities of enterprises to enhance sustainable competitiveness (Barney, 1991). In particular, knowledge is a key resource, and knowledge management practices greatly affect enterprise SDA (Abbas and Sağsan, 2019). Meanwhile, organizational learning with knowledge acquisition as the goal can also enhance the SDA of enterprises (Zhang and Zhu, 2019). Therefore, it is necessary to explore the important role of knowledge resources in the sustainable development of service-oriented manufacturing enterprises.

In fact, in practice, when implementing a service-oriented strategy, enterprises often have manufacturing advantages but lack the accumulation of service resources (Feng *et al.*, 2020), and their knowledge resources are limited. According to resource dependence theory, an enterprise's own knowledge reserve is relatively small; indeed, it cannot fully meet the needs of transformation and upgrading activities, and it is difficult to use such a knowledge reserve to achieve sustainable development (Yu *et al.*, 2021). However, rich knowledge resources maybe more conducive to the design and transmission of such service provision (Akasaka *et al.*, 2012). Thus, it is necessary to search external knowledge resources. As an effective means for service-oriented manufacturing enterprises for obtaining external knowledge, boundary-spanning search (BS) refers to the behavior of enterprises to search for external knowledge resources across organizational boundaries (Grimpe and Sofka, 2009). BS can overcome the limitation of knowledge resources (Jung and Lee, 2016) and provide sufficient support. It is the key to enabling enterprises to maintain SDA in complex environments. In the research on BS, scholars mostly focus on the impact of BS on performance and innovation (Amit and Zott, 2012; Yu *et al.*, 2019), and there is insufficient discussion on the effects of BS and enterprise abilities, especially SDA. Therefore, the introduction of BS to explore external knowledge resources and the SDA of service-oriented manufacturing enterprises has stronger theoretical and practical significance.

Although existing studies have clarified the important role of BS in enterprise growth, the process by which BS can promote the SDA of service-oriented manufacturing enterprises has not been identified. According to knowledge management theory, in order for knowledge to enhance the ability of enterprises, this process requires the acquisition, transformation, absorption and integration of knowledge resources, which are affected by many factors (Li and Zhang, 2010). On the one hand, the knowledge resources obtained via BS are static and disorderly (Shao *et al.*, 2021). This requires further integration of knowledge resources that can be transformed into business value (Dahiyat, 2015). Li-Ying *et al.* (2014) defined knowledge integration (KI) as the behavior of identifying, analyzing, sharing and integrating knowledge through formal processes or structure. Therefore, KI can provide new ideas regarding the impact of BS on SDA. Therefore, this paper takes KI as the starting point to explore the internal mechanism between BS and the SDA of service-oriented manufacturing enterprises. On the other hand, BS is affected by resource dependence, and the process of KI is

affected by previous methods and experience (Obeso and Sarabia, 2018). It will rely on the usual processing flow, rigid knowledge and experience to solve problems, that is, the knowledge inertia (KI-2) phenomenon (Liao, 2002). Although existing studies have examined the phenomenon of KI-2, there is no empirical evidence about its regulatory impact on the relationship between BS and KI. Therefore, this paper introduces KI-2 and discusses its moderating impact on the relationship between BS and KI.

Centered on the resource-based view, this paper analyzes the SDA of service-oriented manufacturing enterprises from the perspective of resources. Second, according to the theory of resource dependence, this paper analyzes the impact of the boundary-spanning acquisition of external knowledge resources on the SDA of service-oriented manufacturing enterprises. In addition, this paper introduces KI and KI-2 from the perspective of knowledge management and reveals its mechanism in detail.

2. Theoretical development and hypotheses

2.1 Boundary-spanning search (BS) and sustainable development ability (SDA)

Chen *et al.* (2018) believe that knowledge resources are important information resources that affect the SDA of enterprises. Based on the organizational search theory, enterprises can adjust and construct their internal knowledge bases by acquiring external knowledge to avoid obstacles such as organizational rigidity and ability traps to solve problems and find opportunities (Wang and Wei, 2017; Han *et al.*, 2020). Especially for service-oriented manufacturing enterprises, the key to achieving service-oriented transformation lies in the need for a wide range of product and service integration knowledge resources. Acquiring external knowledge resources can provide sufficient resources for product and service integration activities and can develop or acquire new abilities to obtain economic benefits and social benefits and improve environmental benefits to enhance enterprise SDA (Seclen-Luna *et al.*, 2021). As an act of searching external knowledge resources, BS can break the boundaries between products and services, enterprises and individuals, and industries, provide a new way for enterprises to search and gain knowledge resources from multiple sources, help inject new services into enterprises, promote product and service innovation, and improve internal operation efficiency, thereby maximizing value for customers (Liu *et al.*, 2022; Shin *et al.*, 2022). This helps to enhance the SDA of enterprises to achieve business objectives and maintain competitive advantage (Su *et al.*, 2018).

For service-oriented manufacturing enterprises, improving product technology and obtaining customers and market resources are key measures in the process of enterprise development. Therefore, based on Van Osch and Steinfield (2016), this paper divides BS into two groups: boundary-spanning search for technical knowledge (BSTK) and boundary-spanning search for market knowledge (BSMK). BSTK is an activity performed by enterprises seeking technical knowledge related to products and services to make up for gaps in their internal technical knowledge and gain technical support for enterprise development. On the one hand, the technical knowledge about R&D, product production and services acquired by enterprises through BSTK can expand the original knowledge base, fill the technology gap to enhance the enterprise's technical abilities. This provides strong support for enterprises to upgrade technology, improve products and services and supply to improve the value of customers' consumption experience (Zhu *et al.*, 2016) and enable enterprises to generate more income, so as to occupy a larger market share and improve the competitive position and SDA. On the other hand, BSTK has strengthened their innovation. Yu *et al.* (2019) believe that the heterogeneity of the knowledge acquired via BS for technological knowledge brings advanced technological and innovative ideas to enterprises, helping them break through their own knowledge reserves and promote technological innovation. This will help enterprises launch new products and high value-added services to better meet the needs

of their customers, increase their scale (Wu *et al.*, 2021), enhance the breadth of their development prospects, and improve their SDA.

BSMK is the behavior by which service-oriented manufacturing enterprises acquire market knowledge from customers, partners, suppliers or other subjects, aiming to discover more new product designs, blue ocean markets and new business models to meet customer needs. It is very important for enterprises to meet customer needs through service and provide customers with product and service portfolios that exceed the value of existing products to improve the SDA of enterprises (Shin *et al.*, 2022). On the one hand, to meet customer needs through BSMK, enterprises not only seek to understand and track customers' attitudes and opinions from a broad and accurate market perspective (Tarazona Montoya *et al.*, 2020), timely identify and explore customers' needs, preferences, pain points and potential needs in the market (Zhang *et al.*, 2017) but also, expand the customer base, improve market share (Xiong *et al.*, 2020) and further expand the scale, improve profitability and promote SDA. On the other hand, enterprises acquire the knowledge of competitors and partners through BSMK to improve their own management mode, incentive mode and decision-making path (Xiao and Zhu, 2016). This will help enterprises carry out management innovation and organizational innovation, and enhance the flexibility and anti-risk ability of enterprises. At the same time, acquiring multiagent knowledge resources can stimulate the generation of new ideas and the know-how of enterprises to update the enterprise ability (Peris-Ortiz *et al.*, 2018) and further cultivate the SDA of enterprises. Therefore, this paper hypothesizes the following:

H1a, H1b. The SDA of service-oriented manufacturing enterprises is positively affected by BSTK (BSMK).

2.2 Boundary-spanning search (BS) and knowledge integration (KI)

Existing research points out that rich knowledge resources can enhance the potential of KI (West *et al.*, 2014). Laursen and Salter (2006) believe that by absorbing external knowledge through BS, enterprises can adjust and build their knowledge systems and internal knowledge, which strengthen their KI behavior. This is reflected in the following three aspects. First, BS promotes the digestion of knowledge in the KI process. Through the BSTK, enterprises increase the frequency of their exchange with external technical knowledge and more clearly define the standards of technical knowledge to enhance their ability to recognize technical knowledge, narrow the gap between them and the outside in terms of technical knowledge and clarify the standards of technical knowledge. This will help enterprises absorb and transform knowledge (Huang and Li, 2017). Second, BS promotes knowledge flow in the KI process. Acquiring advanced technical knowledge based on BSTK provides technical support for enterprises to establish an effective knowledge sharing platform to promote knowledge exchange and sharing between internal personnel and organizations (Men *et al.*, 2019). Third, the BS encourages enterprises to interact with multiple parties in the KI process. By adopting BSMK behavior, enterprises can virtually strengthen the interaction between enterprises, customers and suppliers, and strengthen the connection between enterprises and external markets (Xu *et al.*, 2017). This not only improves the flow rate of external knowledge but also makes the process of interacting with the market easy for internal personnel to understand and trust, thus promoting the internalization of knowledge (Liu *et al.*, 2021) and improving the efficiency of KI. This paper hypothesizes the following:

H2a, H2b. BSTK (BSMK) enhances enterprise KI.

2.3 Knowledge integration (KI) and sustainable development ability (SDA)

According to knowledge-based theory, the way in which knowledge is stored and used has a profound impact on the survival and development of enterprises, especially in terms of the

integration and coordination of knowledge. Indeed, [Verona and Ravasi \(2003\)](#) believe that KI is the internal basis allowing enterprises to improve their SDA. First, KI not only improves the utilization rate of knowledge and reduces knowledge redundancy in the process of product and service provision but also further promotes the generation of new knowledge ([Maenpaa et al., 2016](#)) through member creativity and brings continuous innovation abilities to enterprises. Additionally, the process of KI deepens the level of understanding and close communication of knowledge among the internal members of an enterprise and enhances the efficient cooperation between them ([Zahra et al., 2000](#)). The knowledge produced is effective and of a high quality, that is, knowledge that can accurately capture the needs of customers, enabling the enterprise to develop products and high value-added services that meet the needs of its customers, increase profits, expand its market scale and improve its SDA.

However, the effect of KI is not a linear feature without change because as KI deepens, the cost of KI gradually increases ([Wu et al., 2020](#)). That is, enterprises need to devote more human resources, finances and time to ensure that more KI activities can be carried out smoothly ([Tsai et al., 2015](#)). Thus, resources and energy devoted to important activities such as product manufacturing and service provision are diverted, and the expansion of scale and market share are delayed. In addition, previous studies show that excessive KI makes it easy for enterprises to fall into and rely on organizational rigidity ([Lee et al., 2017](#)). This is because excessive KI leads enterprises to give too much attention to and make direct use of familiar knowledge fields, with little improvement in terms of new knowledge. In this case, enterprises are eventually hindered from discovering new technologies, and their ability to identify market opportunities is reduced, thus hindering service-oriented manufacturing, which is not conducive to the improvement of their SDA. Therefore, this paper hypothesizes the following:

- H3.* The KI and SDA of service-oriented manufacturing enterprises have an inverted U-shaped relationship.

2.4 Mediating effect of knowledge integration (KI)

Academia has paid attention to the intermediary role of KI. For example, [Zhang et al. \(2021\)](#) found that after looking for knowledge resources, one must take the intermediate step of KI to achieve breakthrough innovation. For service-oriented manufacturing enterprises, in order to provide new products and services that satisfy customers, they need to effectively integrate new knowledge acquired externally with internal knowledge ([Liu and Yang, 2019](#)). Specifically, first, static knowledge resources acquired through BSTK and BSMK cannot be directly transformed into commercial value, which requires KI. Since knowledge resources involve different subjects and contain complex contents, they are not systematic and cannot be directly utilized according to needs and environmental changes ([Meng and Yang, 2018](#)). Only by understanding and absorbing complex and dispersed knowledge can enterprises restructure knowledge resources, enhance competitive advantages and improve SDA ([Feng and Chen, 2015](#)). Second, KI can integrate the acquired external knowledge with its own knowledge, form a unique knowledge system, give play to the value of knowledge ([Dahiyat, 2015](#)) and translate this into the improvement of enterprise SDA. Thus, this paper hypothesizes the following:

- H4a, H4b.* KI has a mediating effect on the relationship between BSTK (BSMK) and the SDA of service-oriented manufacturing enterprises.

2.5 Moderating effect of knowledge inertia (KI-2)

Enterprises can save costs by relying on past internal knowledge and experience to solve problems. For example, enterprises do not need to spend more time and money to find new

solutions (Liao, 2002). According to the theory of organizational inertia, if enterprises continue to follow inertia and past experience, they will stagnate and hinder their own development. Yu *et al.* (2020) believe that acquired external knowledge will lead to confusion and complexity of information due to KI-2, resulting in redundancy of knowledge resources. In this regard, the new knowledge obtained by BS is more likely to become the burden of enterprise resources, increasing the difficulty, cost and time of KI (Zhou *et al.*, 2014). In addition, KI-2 leads enterprises to integrate old knowledge processing methods and communication mechanisms (Liao, 2002), which reduces the flow of new knowledge, hinders knowledge exchange and sharing among enterprise members and departments and is not conducive to KI. Therefore, a relatively large KI-2 means that enterprises are greatly affected by past experience, process and thinking, and it is difficult to accept external heterogeneous knowledge obtained by BS, which hinders KI. This paper assumes the following:

H5a, H5b. KI-2 has a negative moderating effect on the relationship between BSTK (BSMK) and KI.

Based on these concepts, we propose the theoretical model of this paper, as shown in Figure 1.

3. Methods

3.1 Data collection and processing

Service-oriented manufacturing enterprises are taken as the research object of this paper. The enterprise selection process was as follows: first, the service-oriented manufacturing demonstration enterprises announced by the Ministry of Industry and Information Technology of the People’s Republic of China were selected; second, because the Yangtze River Delta region is a highly concentrated area of service-oriented manufacturing enterprises and an important area of regional service, the service-oriented manufacturing demonstration enterprises announced by the Ministry of Industry and Information Technology of provinces in the Yangtze River Delta region were selected; third, to obtain enterprise financial data, the non-listed companies and companies with financial status and delisting risk warnings in the first and second steps were excluded. Finally, Cohen’ Postulates of Statistical Power was used in this study, and combined with the maximum number of independent variables of this study was three, it was concluded that the study needed 37 observations to a statistical power of 80% for detecting R^2 values of at least 0.25 (with a 5% probability of error) (Hair *et al.*, 2017). This paper selects a total of 110 according to the sample demand, which conforms to the criteria for determining the sample size. 110 service-oriented manufacturing

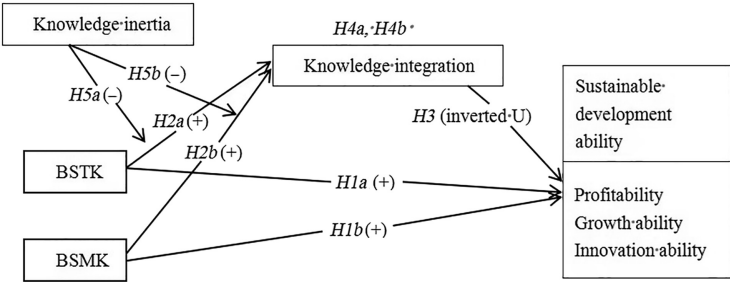


Figure 1.
Conceptual model

enterprises were selected. Among them, the specific situation of 110 sample enterprises in the provinces is shown in Table 1.

This paper collects data through a questionnaire survey and enterprise annual reports. The data of dependent variables and control variables are obtained from the 2017–2019 annual reports of enterprises, and the data of other variables are collected by expert scoring questionnaires. The data sources of the questionnaire are as follows: first, the meaning of each variable was clarified by carefully examining the literature. Information about BS, KI and KI-2 of 110 enterprises was collected from the annual report and the official website of the enterprise. Approximately 10,200 words of enterprise variable information were sorted out. Second, to obtain comprehensive and objective data, three experts from each enterprise completed the questionnaire. Then, dividing the answers of three interviewees of an enterprise into a group helped form a triangular verification to determine whether the data corroborate each other. Among them, the research direction of the interviewed experts was required to be related to manufacturing service, knowledge management, innovation strategy and have a certain knowledge base to deeply understand the content of the item. Finally, 330 questionnaires were obtained. The descriptive statistics of 110 samples and experts are shown in Table 1.

3.2 Variable setting

Variable measurement mainly drew on mature scales used by scholars, considered the characteristics of the research purpose, and finally determined the measurement items based on interviews professionals. Measurement items were measured by a five-point Likert scale (1 means strongly disagree, 5 means strongly agree).

Item	Classification	Ratio	Item	Classification	Ratio
Size	Below 1,000	23.38	provinces	Zhejiang	39.09
	1,000–2,000	22.08		Jiangsu	25.45
	2,000–5,000	25.97		Anhui	14.54
	5,000–10,000	10.39		Shanghai	3.63
	Over 10,000	18.18		Guangdong	2.73
Industry	Instrument	5.45	Years in job	Tianjin	1.82
	Automobile	5.45		Liaoning	1.82
	Furniture	8.18		Shandong	1.82
	Special equipment	15.45		Jiangxi	1.82
	General machinery	10.91		Fujian	0.91
	Rubber and plastic products	1.82		Chongqing	0.91
	Other manufacturing	15.45		Xian	0.91
	Fabricated metal	2.73		Xinjiang	0.91
	Wine, beverages and refined tea	3.64		Henan	0.91
	Textile and apparel, clothing	4.55		Hubei	0.91
	Electrical machinery and equipment	14.54		Sichuan	0.91
	Chemical raw materials and chemicals	2.73		Gansu	0.91
	Leather, fur, feathers and their products and footwear	1.82		0–3	47.88
	Cultural and educational supplies, industrial, sports and entertainment	1.82		3–5	22.12
	Computers, communications and other electronic equipment	3.64		5–10	12.73
	Black metal smelting and rolling processing	1.82		Over 10	17.27
Education	Bachelor	14.24			
	Master	47.27			
	Doctor	38.49			

Table 1.
Sample's
characteristics
distribution

(1) Dependent variable

The SDA of service-oriented manufacturing enterprises was measured based on the relevant measurement indicators of SDA proposed by [Su and Yu \(2018\)](#) and [Su et al. \(2018\)](#). It was rooted in three aspects: profitability (the net profit margin and return on total assets), growth ability (the growth rates of net assets and total assets) and innovation ability (innovation investment intensity and the proportion of R&D personnel).

(2) Independent variable

The scale proposed by [Martini et al. \(2017\)](#) and [Zhu et al. \(2016\)](#) was used to measure the two dimensions of BS variables, namely, BSTK and BSMK. Therefore, this paper constructs eight items to measure three indicators: knowledge cognition, search degree and search channel.

(3) Mediating variable

Based on the research of [Alegre et al. \(2013\)](#), this paper made appropriate adjustments based on the characteristics of the research topic and measured them based on two indicators, i.e. knowledge transfer and knowledge application, using four items.

(4) Moderating variable

For the moderating variable KI-2, this paper referred to the scale design proposed by [Xie et al. \(2016\)](#), measured based on two indicators, i.e. dependence and hindrance, and used four items.

(5) Control variables.

The SDA of enterprises is affected by many factors. In this study, the following variables were selected as control variables. Enterprise size (ES): expressed by the number of employees; Enterprise years (EY): the number of years since the establishment of the enterprise; Financial leverage (Lev): measured by asset liability ratio; EBITDA: expressed by the sum of EBIT and enterprise depreciation and amortization. In addition, the industry type (IT) of the enterprise is also included as the control variable.

3.3 Regression model

This study constructs a multiple regression model to test hypotheses. First, this study analyzes the impact of BSTK and BSMK on KI and SDA, respectively, and select KI and SDA as the explained variables to test the regression models of [H1a](#), [H1b](#), [H2a](#) and [H2b](#) as follows:

$$SDA_i/KI = \lambda_0 + \lambda_1 BSTK + \lambda_2 BSMK + \lambda_3 Controls + \varepsilon_i \tag{1}$$

The subscript i of the variable represents the enterprise. The explanatory variables are SDA and KI. The explanatory variables are BSTK and BSMK. The control variables include ES, EY, Lev (Financial coverage), EBITDA and IT. λ_0 is a constant term and ε_i is a random perturbation term.

Second, this study analyzes the impact of KI on SDA, select SDA as the explained variable to test [H3](#), and the regression model is,

$$SDA_i = \lambda_0 + \lambda_1 KI + \lambda_2 KI \times KI + \lambda_3 Controls + \varepsilon_i \tag{2}$$

Third, this study analyzes the mediating effect of KI. On the basis of [formula \(1\)](#), combine the following [formula \(3\)](#) to judge the mediating effect of KI to test [H4a](#) and [H4b](#). The regression model is,

$$SDA_i = \lambda_0 + \lambda_1BSTK + \lambda_2BSMK + \lambda_3KI + \lambda_4Controls + \varepsilon_i \tag{3}$$

Finally, this study analyzes the moderating effect of KI-2, construct the interaction term of BSTK, BSMK and KI-2, and take KI as the explained variable to test [H5a](#) and [H5b](#). The regression model is,

$$KI = \lambda_0 + \lambda_1BSTK + \lambda_2BSMK + \lambda_3KI - 2 + \lambda_4BSTK \times KI - 2 + \lambda_5BSMK \times KI - 2 + \lambda_6Controls + \varepsilon_i \tag{4}$$

3.4 Assessment of the measures: reliability, validity and common method variance

As shown in [Table 2](#), first, the final variables were calculated as the mean of their indicators, and the α of each variable was higher than 0.7, indicating that the internal consistency of the scale was high. Second, the KMO value was 0.846, indicating that further analysis is possible. $X^2/df = 1.213$, i.e., less than 3; RMSEA = 0.044, i.e. less than 0.08. The indicators met the standard, indicating that the model had a good degree of fitting. The factor loading value of each item exceeded 0.65, indicating good convergent validity. The AVE variable value was larger than 0.5, and the CR value was greater than 0.7. The relevant statistical data of the variables are shown in [Table 3](#). In addition, this paper uses variable inflation factor (VIFs) to test whether there is multicollinearity between variables. The results show that the maximum value of the variable VIF is 1.462, which is less than 10; that is, there is no serious multicollinearity problem.

Variable	Measurement index	Load	Reliability and validity index
BSTK	R&D of technologies and new products in the industry at home and abroad	0.876	$\alpha = 0.860$ AVE = 0.615 CR = 0.864
	The technological development of other industries with a potential impact	0.826	
	Platform technology for jointly developing products and services with universities and scientific research institutes	0.681	
	Attendance at international exhibitions and seminars	0.739	
BSMK	Changes in customer demand and preferences for products and services	0.829	$\alpha = 0.875$ AVE = 0.638 CR = 0.876
	The use of customer and feedback data on products and services	0.769	
	Customer innovation activities in product production and service process	0.814	
	Product R&D and service provision of competitors (or partners)	0.781	
KI	The establishment of relevant platforms or systems to realize knowledge transfer	0.845	$\alpha = 0.893$ AVE = 0.680 CR = 0.895
	The sharing of acquired knowledge among members	0.827	
	The establishment of a communication mechanism between members and departments for knowledge exchange	0.869	
	New knowledge leads to new products and services to satisfy customers demands	0.753	
KI-2	Deal with new problems with old knowledge	0.702	$\alpha = 0.821$ AVE = 0.541 CR = 0.824
	The learning process depends on past knowledge and experience	0.678	
	Learning new knowledge is influenced by old knowledge and experience	0.747	
	Gain new knowledge from old	0.808	

Table 2.
Reliability and validity tests

Table 3.
Descriptive statistics
and
correlations (N = 110)

	EBITDA	Lev	ES	EY	IT	BSTK	BSMK	KI	KI-2	SDA
EBITDA	1									
Lev	0.347**	1								
ES	0.524**	0.589**	1							
EY	0.168	0.281**	0.160	1						
IT	0.048	-0.034	0.014	-0.063	1					
BSTK	0.099	-0.114	0.093	-0.015	-0.028	1				
BSMK	0.064	-0.187	0.098	-0.135	-0.053	0.420**	1			
KI	-0.013	-0.161	0.079	-0.039	0.069	0.450**	-0.101	1		
KI-2	0.025	0.047	-0.085	0.005	-0.168	-0.114	0.418**	-0.241*	1	
SAD	-0.084	-0.302**	-0.089	-0.174	-0.168	0.384**	0.454**	0.400**	-0.207*	1
Mean	2,405,933,587	0.399	5709.6	20.11	6.67	4.10	3.94	3.90	2.93	0.0006
SD	6,653,500,789	0.178	11063.3	7.27	4.87	0.52	0.51	0.51	0.45	0.75
Note(s): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$										

4. Empirical results

4.1 Direct effect test

As shown in Table 4, models 1 to 6 are built to test H1a, H1b, H2a, H2b and H3. Model 1 is the control variable model. Model 2 and model 3 add two independent variables, BSTK and BSMK, to model 1. The analysis results of model 2 show that BSTK is positively correlated with SDA ($\beta = 0.343, p < 0.001$), indicating that BSTK positively affects SDA. Thus, H1a is supported. Similarly, the model 3 results show that BSMK is positively correlated with SDA ($\beta = 0.394, p < 0.001$), indicating a BSMK-positive collision on SDA. Thus, H1b is supported. However, by comparing the influence coefficients of the two, it is found that BSMK has a greater influence on SDA than BSTK.

Based on model 1 and model 4, KI and the square term of KI are considered. For SDA, KI is of paramount value ($\beta = 0.358, p < 0.01$). Additionally, the correlation between the square term of KI and SDA is negative. However, it was not significant ($\beta = -0.019, p > 0.05$). Thus, H3 is not supported. It can be seen that KI has a positive effect on SDA, while the inverted U-shaped effect has not yet been shown.

In addition, models 5 and 6 take BSTK and BSMK as independent variables, respectively, to test their influence on KI. It can be seen from the description in Table 4 that the regression coefficients of BSTK and BSMK are 0.420 and 0.385, respectively, and they are significant at the $p < 0.001$ level, indicating that BSTK and BSMK have a direct positive effect on KI. H2a and H2b are confirmed.

4.2 Mediating effect test

Models 1–6 in Table 5 are constructed to test H4a and H4b. BSTK and BSMK are used as independent variables to test the mediating effect of KI. Model 1 shows that BSTK and KI are positively and significantly correlated ($\beta = 0.420, p < 0.001$). Additionally, model 2 indicates that the positive effect of BSTK on SDA is of great significance ($\beta = 0.343, p < 0.001$). Model 3 introduces KI into the regression model on the basis of model 2 and shows that BSTK and KI exert influence on SDA ($\beta = 0.230, p < 0.05$; $\beta = 0.269, p < 0.01$). Compared with the results from model 2, the absolute value of the regression coefficient of BSTK significantly decreases, indicating that there is a mediating effect. Therefore, H4a is supported.

With BSMK as the independent variable, the results show that BSMK exerts a significant positive impact on both KI ($\beta = 0.385, p < 0.001$) and SDA ($\beta = 0.394, p < 0.001$). Rooted in model 5, a mediating variable is introduced into construct model 6, which indicates that the coefficient of BSMK is positive ($\beta = 0.294, p < 0.01$) and that the coefficient of KI is also

Variable	SDA			KI		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
EBITDA	−0.009	−0.039	−0.029	0.013	−0.100	−0.084
Lev	−0.364**	−0.271*	−0.224*	−0.250*	−0.202	−0.178
ES	0.150	0.079	0.033	0.043	0.208	0.180
EY	−0.106	−0.109	−0.068	−0.112	0.013	0.054
IT	−0.189*	−0.174*	−0.158	−0.213*	0.077	0.088
BSTK		0.343***			0.420***	
BSMK			0.394***			0.385***
KI				0.358**		
KI × KI				−0.019		
R ²	0.147	0.258	0.285	0.272	0.244	0.209
Adjusted R ²	0.106	0.215	0.243	0.222	0.200	0.163
F	3.598**	5.970***	6.826***	5.451***	5.536***	4.534***
Note(s): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$						

Table 4.
Direct effect regression
analysis

Table 5.
Mediating effect of
knowledge integration

Variable	KI		SDA		KI		SDA	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 6	Model 6
EBITDA	−0.100	−0.039	−0.012	−0.084	−0.029	−0.008		
Lev	−0.202	−0.271 [*]	−0.217 [*]	−0.178	−0.224 [*]	−0.178		
ES	0.208	0.079	0.023	0.180	0.033	−0.014		
EY	0.013	−0.109	−0.113	0.054	−0.068	−0.082		
IT	0.077	−0.174 [*]	−0.195 [*]	0.088	−0.158	−0.181 [*]		
BSTK	0.420 ^{***}	0.343 ^{***}	0.230 [*]					
BSMK				0.385 ^{***}	0.394 ^{***}	0.294 ^{**}		
KI			0.269 ^{**}			0.259 ^{**}		
R ²	0.244	0.258	0.313	0.209	0.285	0.338		
Adjusted R ²	0.200	0.215	0.266	0.163	0.243	0.292		
F	5.536 ^{***}	5.970 ^{***}	6.631 ^{***}	4.534 ^{***}	6.826 ^{***}	7.426 ^{***}		

Note(s): ^{*}*p* < 0.05, ^{**}*p* < 0.01, ^{***}*p* < 0.001

positive and significant ($\beta = 0.259, p < 0.01$). Observing model 5 and model 6, according to the comparison, it is found that the absolute value decreases, and the significance decreases, which proves that there is a partial mediating effect. Therefore, H4b is supported.

The mediating KI effect is preliminarily verified. Based on this, the indirect effect of KI on its relationship is further verified by a bootstrap test. The results are shown in Table 6. The bootstrap 95% confidence intervals of the mediating effect of KI, with BSTK and BSMK as the independent variables, are (0.039, 0.305) and (0.022, 0.310), respectively, excluding 0. These results indicate that the BSTK and BSMK can not only have a more direct impact on SDA but also have a positive impact on SDA through the mediating effect. The results also further confirm that BSTK and BSMK have mediating effects on SDA. KI is the mediating variable. Table 6 shows that the direct impact value of BSTK on SDA is 0.332, and the indirect impact value is 0.163. The direct influence value of the BSMK on SDA is 0.435, and the indirect influence value is 0.147. The direct effect of BSTK or BSMK on SDA is stronger than the indirect effect of KI on SDA.

4.3 Moderating effect test

In Table 7, models 1–6 are constructed to test H5a and H5b. The model 3 and model 5 results show that BSTK and BSMK are independent variables. After adding the KI-2 moderating variable and the interaction term, the regression coefficients of the interaction term, −0.182 and −0.238, respectively, are negative and significant at $p < 0.05$. Therefore, KI-2 has a negative moderating influence between BSTK and KI relationships and between BSMK and KI. Hence, H5a and H5b are supported.

Based on the operation suggestions of Aiken and West (1991), the simple slope of the saturated regression equation is used to draw the relationship between BSTK, BSMK and KI at the level of one standard deviation above and below the mean value KI-2. In Figure 2, under low KI-2, the straight line changes from flat to step, and enhancing BSTK has a stronger positive impact on KI. These results indicate that KI-2 negatively moderates the impact of BSTK on KI. Figure 3 is similar to Figure 2. Under low KI-2, BSMK has a stronger impact on KI. With an increase in KI-2, the impact of BSMK on KI becomes weaker, indicating that KI-2 negatively moderates the impact of BSMK on KI.

Table 6.
Bootstrap mediating
effect test

Variable	Total effect	Direct effect	Indirect effect	BootSE	BootLLCI	BootULCI
BSTK	0.495	0.332	0.163	0.067	0.039	0.305
BSMK	0.582	0.435	0.147	0.075	0.022	0.310

Variable	Sustainable development ability				
	Model 1	Model 2	KI Model 3	Model 5	Model 6
EBITDA	−0.064	−0.082	−0.073	−0.065	−0.053
Lev	−0.315*	−0.183	−0.150	−0.159	−0.127
ES	0.294*	0.176	0.156	0.147	0.115
EA	0.017	0.008	−0.007	0.048	0.017
IT	0.059	0.049	0.053	0.058	0.061
BSTK		0.404***	0.372***		
BSMK				0.370***	0.358***
KI-2		−0.162	−0.179*	−0.173	−0.194*
BSTK × KI-2			−0.182*		
BSMK × KI-2					−0.238**
R ²	0.078	0.268	0.299	0.237	0.181
Adjusted R ²	0.034	0.218	0.244	0.185	0.040
F	1.763	5.343***	5.386***	4.524***	8.269***

Note(s): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7.
The moderating effect of knowledge inertia

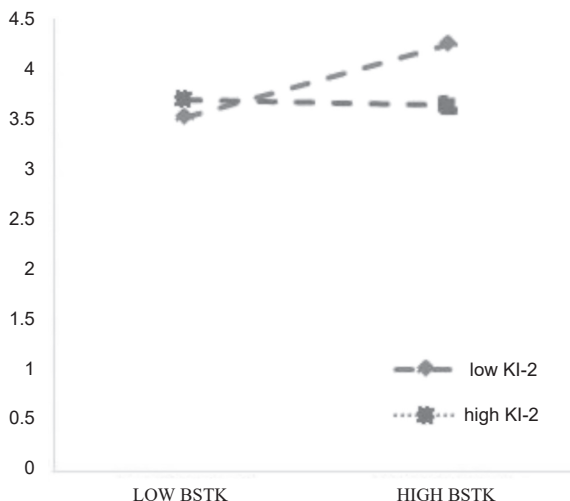


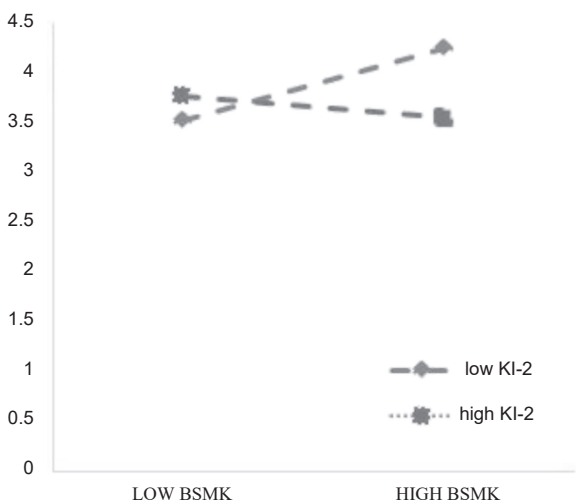
Figure 2.
The moderating effect of knowledge inertia on BSTK and knowledge integration

5. Conclusion

This research shows that acquiring external knowledge across organizational boundaries and integrating knowledge in service-oriented manufacturing enterprises is the key to enhancing SDA. This study analyzes the data of service-oriented manufacturing enterprises and draw the following main conclusions. First, both the external BS and internal KI promote SDA. However, in the context of China, the development of China's service-oriented manufacturing industry is still in its initial stage. KI's impact on SDA has not yet reached the highest point of SDA improvement, nor has it reached the inflection point, showing no inverted U-shaped impact. Second, BS influences SDA through KI, which acts as a mediator in this process. However, by comparing the direct and indirect effects of BSTK and BSMK on SDA. It is found that the direct effect is stronger than the indirect effect through KI. Third, the influence of BS on KI will be negatively regulated by KI-2. The findings of this study contribute to the literature in three aspects.

Figure 3.

The moderating effect of knowledge inertia on BSMK and knowledge integration



First, previous studies have focused on the relationship between BS and enterprise innovation behavior, ignoring the wider impact of variables. This paper introduced the SDA variable of service-oriented manufacturing enterprises into the existing research to explore the mechanism of BS on SDA, which is an in-depth expansion and supplement to the research on SDA of service-oriented manufacturing enterprises, and found that the research conclusions are consistent with the content of the resource-based view. Second, from the perspective of combining internal and external perspectives, this study introduces the mediator variable of KI to reveal the internal mechanism of the impact of BS on SDA, opens the “black box” between the two, and makes up for the problems that have not been discussed in depth in previous studies on the process of BS. Third, in contrast to previous studies on the direct relationship between BS and KI, this paper, based on knowledge management theory, reveals the moderating effect of KI-2 on the relationship between BS and KI and finds that the acquisition of new knowledge will conflict with the dependence on internal knowledge practices; that is, internal rigid knowledge experience will repel new knowledge, thus damaging the knowledge flow rate and knowledge sharing process and inhibiting KI behavior. This is a further expansion of BS.

5.1 Managerial implications

There are three main management implications of this study. First, the study found that enterprises can enhance their SDA through the BSTK and BSMK. However, the influence of BSTK and BSMK on SDA is different, and the influence of BSMK on SDA is greater than that of BSTK. Because service-oriented manufacturing enterprises are customer-centric, meeting customer needs is the ultimate goal of enterprises. Therefore, through BSMK, enterprises can more directly obtain customer feedback, demand knowledge resources involved in services. Enterprises should pay more attention to BSMK when adopting BS behavior. Therefore, service-oriented manufacturing enterprises should broaden BS channels and increase external knowledge resources. Enterprises can find more heterogeneous knowledge from multiple subjects (including customers, suppliers, competitors, scientific research institutes, etc.). Enterprises can communicate with each subject through cooperation, incentives and other means. They can also use emerging technologies to quickly and accurately acquire novel and diverse technical knowledge and market knowledge.

Second, we note that KI can not only directly enhance the SDA of enterprises, but also play an intermediary role in the process of BS affecting SDA. We suggest that enterprises improve the internal knowledge flow rate and enhance the understanding and absorption of knowledge by employees. Specifically, enterprises can use big data to achieve the digital empowerment of KI and can use a variety of technologies to simplify the KI process (Zhang *et al.*, 2020) and filter and integrate knowledge features based on data to make the knowledge system more orderly (Tang *et al.*, 2020), which will help improve the efficiency of KI and improve the SDA of enterprises.

Third, we suggest that managers establish dynamic management systems and processes, break with conventions and accept new ways. In addition, enterprises provide more training and external communication and learning opportunities to break through KI-2 and stimulate new knowledge of employees.

5.2 Limitations and future research

Regarding the limitations of the study, first, the scope of research objects is small. The research object of this paper is the service-oriented manufacturing industry, which is different from other types of enterprises such as service enterprises. Whether the conceptual model is applicable to other enterprises needs further testing. Future research should expand the scope of research objects. In addition, this study does not consider the impact of the external environment on BS and SDA. Follow-up research can consider the impact of the external environment such as market fluctuation, the digital environment and technological change, and obtain more meaningful research results.

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